

# Machine Health Intelligence (MHI)

AI-Powered Predictive Maintenance Report

## Machine Parameters

**Machine Type:** L  
**Air Temperature:** 300.0 K  
**Process Temp:** 310.0 K  
**Rotational Speed:** 1500 RPM  
**Torque:** 100.0 Nm  
**Tool Wear:** 299 min

## Prediction Summary

Failure Probability : 90.89%  
Health Score : 9.11%  
Risk Level : Critical  
Estimated Downtime : 6–8 Hours  
Estimated Cost : ■30,000+

## Failure Causes

- High Torque
- Excessive Tool Wear

## Maintenance Recommendations

1. Stop machine immediately.
2. Perform complete mechanical inspection.
3. Replace worn cutting tool.
4. Check and service the cooling system.
5. Inspect spindle bearings.
6. Reduce operating load before restart.

Generated on 2026-07-14 16:10:19